

- 4) What Greek letters are used to represent the Type I and II error rates.

- 5) What levels are conventionally used for significance testing?

- 6) When is it valid to use a one-tailed test? What is the advantage of a one-tailed test? Give an example of a null hypothesis that would be tested by a one-tailed test.

- 7) Distinguish between probability value and significance level?

- 8) The following are 12 determinations of the melting point of a compound (in degrees centigrade) made by an analyst, the true melting point being 165°C . Would you conclude from these data that her determination is free from bias?
 164.4, 161.4, 169.7, 162.2, 163.9, 168.5, 162.1, 163.4, 160.9, 162.9, 160.8, 167.7.

31.7.2 Comparison of Two Univariate Normal Distributions

Let the distribution of x in each of two populations be normal. Suppose the mean and the standard deviation of x for one population are μ_1 and σ_1 , for the other they are μ_2 and σ_2 , respectively. Suppose further that $x_{11}, x_{12}, \dots, x_{1n}$ are a random sample from the first distribution, and $x_{21}, x_{22}, \dots, x_{2n}$ are a random sample for the second set. The first set of observation is also supposed to be independent of the second set. Then $\bar{x}_1 = \sum_j^{n_1} x_{1j} / n_1$ and $s_1' = \sqrt{\sum_j^{n_1} (x_{1j} - \bar{x}_1)^2 / (n_1 - 1)}$ are the

mean and standard deviation of x in the first sample, and $\bar{x}_2 = \sum_{j=1}^{n_2} x_{2j} / n_2$,

$s_2' = \sqrt{\sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)^2 / (n_2 - 1)}$ are the corresponding statistics of the second sample.

Case I: μ_1, μ_2 unknown but σ_1, σ_2 known

In this case, one may be concerned with a comparison between the population means. One may have to test the hypothesis that μ_1 and μ_2 differ by a specified quantity, say $H_0: \mu_1 - \mu_2 = \xi_0$, or one may like to obtain confidence limits for the difference $\mu_1 - \mu_2$.

It may be seen that $\bar{x}_1 - \bar{x}_2$, being a linear function of normal variables, is itself normally distributed. It has mean $E(\bar{x}_1 - \bar{x}_2) = E(\bar{x}_1) - E(\bar{x}_2) = \mu_1 - \mu_2$ and variance $\text{var}(\bar{x}_1 - \bar{x}_2) = \text{var}(\bar{x}_1) + \text{var}(\bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$, and the covariance term being zero since $\bar{x}_1$ and $\bar{x}_2$ are independent. As such $\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})^{1/2}}$ is

distributed as a standard normal variable. To test $H_0: \mu_1 - \mu_2 = \xi_0$, we make use of the statistic $\frac{(\bar{x}_1 - \bar{x}_2) - \xi_0}{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})^{1/2}}$, which is distributed as a standard normal variable

(τ) under H_0 . H_0 is to be rejected on the basis of the given samples if $\tau > \tau_a$ or if $\tau < -\tau_a$, according as the alternative hypothesis in which the experimenter is interested i.e., in $H_1: \mu_1 - \mu_2 > \xi_0$ or $H_1: \mu_1 - \mu_2 < \xi_0$.

On the other and, if the alternative is $H: \mu_1 - \mu_2 \neq \xi_0$, H_0 is to be rejected when $|\tau| > \tau_{a/2}$. In the commonest cases, the null hypothesis will be $H_0: \mu_1 = \mu_2$ for which $\xi_0 = 0$. If the problem is one of interval estimation, then it will be found that following the usual mode of argument that the confidence limits to $\mu_1 = \mu_2$ (with confidence coefficient $1 - \alpha$) are $(\bar{x}_1 - \bar{x}_2) - \tau_{a/2}(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})^{1/2}$ and

$$(\bar{x}_1 - \bar{x}_2) + \tau_{a/2}(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})^{1/2}.$$

Case II: μ_1, μ_2 known but σ_1, σ_2 unknown

Here it may be necessary to test the hypothesis that the ratio of the two unknown standard deviations has a specified value, say $H_0: \sigma_1 / \sigma_2 = \xi_0$, or to

set confidence limits to this ratio. Since $\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / \sigma_1^2$ and $\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / \sigma_2^2$ are independent χ^2 s with n_1 and n_2 degrees of freedom,

$$\text{respectively, } P \left[\frac{1}{F_{\alpha/2; n_2, n_1}} \leq \frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / n_1 \sigma_1^2}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / n_2 \sigma_2^2} \leq \frac{1}{F_{\alpha/2; n_1, n_2}} \right] = 1 - \alpha \quad \text{where}$$

$\frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / n_1 \sigma_1^2}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / n_2 \sigma_2^2}$ is distributed as an F with (n_1, n_2) df. Under the

hypothesis $H_0: \sigma_1 / \sigma_2 = \xi_0$, therefore, $\frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / \sigma_1^2}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / \sigma_2^2} \times \frac{1}{\xi_0^2}$ is an F with df =

(n_1, n_2) . This provides a test for H_0 . When the alternative is $H_1: \sigma_1 / \sigma_2 > \xi_0$, H_0 is to be rejected if for the given samples $F > F_{\alpha; n_1, n_2}$.

If the alternative is $H_1: \sigma_1 / \sigma_2 < \xi_0$, H_0 is to be rejected if for the given samples

$F < F_{1-\alpha; n_1, n_2}$, i.e., if $1/F > F_{\alpha; n_2, n_1}$.

Lastly, when the alternative is $H_1: \sigma_1 / \sigma_2 \neq \xi_0$, H_0 is to be rejected if the samples in hand give either $F < F_{1-\alpha/2; n_1, n_2}$, i.e., $1/F > F_{\alpha/2; n_2, n_1}$, or

$F > F_{\alpha/2; n_2, n_1}$.

The commonest form of the null hypothesis will be $H_0: \sigma_1 = \sigma_2$ for which $\xi_0 = 1$,

$$\text{and hence, } F = \frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / n_1}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / n_2}.$$

For the purpose of setting confidence limits to σ_1 / σ_2 , we see that

$$P \left[\frac{1}{F_{\alpha/2; n_2, n_1}} \leq \frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / n_1 \sigma_1^2}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / n_2 \sigma_2^2} \leq \frac{1}{F_{\alpha/2; n_1, n_2}} \right] = 1 - \alpha$$

$$\text{i.e., } P \left[\frac{1}{F_{\alpha/2; n_2, n_1}} \frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / n_1}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / n_2} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{1}{F_{\alpha/2; n_1, n_2}} \frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / n_1}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / n_2} \right] = 1 - \alpha$$

The confidence limits to $\frac{\sigma_1^2}{\sigma_2^2}$ with confidence coefficient $(1 - \alpha)$ will, therefore,

$$\text{be } \frac{1}{F_{\alpha/2; n_1, n_2}} \frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / n_1}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / n_2} \text{ and } \frac{1}{F_{\alpha/2; n_2, n_1}} \frac{\sum_{j=1}^{n_1} (x_{1j} - \mu_1)^2 / n_1}{\sum_{j=1}^{n_2} (x_{2j} - \mu_2)^2 / n_2}.$$

The corresponding limits to σ_1 / σ_2 will naturally be the positive square roots of these quantities.

Case III: Means and standard deviations all unknown

We shall first consider methods of testing for the difference of the two means and of setting confidence limits to this difference.

For the sake of simplicity, we shall assume that the two unknown standard deviations are equal. Now if σ denotes the common standard deviation, then

$$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{1/2}} \text{ is a standard normal variable, while}$$

$$\frac{(n_1 - 1) s_1'^2 + (n_2 - 1) s_2'^2}{\sigma^2} = \frac{\sum_{j=1}^{n_1} (x_{1j} - \bar{x}_1)^2}{\sigma^2} + \frac{\sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)^2}{\sigma^2}$$

which is the sum of two independent χ^2 s, one with $df = (n_1 - 1)$ and the other with $df = (n_2 - 1)$ and hence is itself a χ^2 with $df = (n_1 + n_2 - 2)$.

Hence, denoting by s'^2 the pooled variance of the two samples, so that

$$s'^2 = \frac{(n_1 - 1) s_1'^2 + (n_2 - 1) s_2'^2}{n_1 + n_2 - 2} \text{ we have}$$

$$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s' \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{1/2}}}{\left\{ \frac{(n_1 - 1) s_1'^2 + (n_2 - 1) s_2'^2}{\sigma^2 (n_1 + n_2 - 2)} \right\}^{1/2}}$$

is a quantity of the form $\frac{\tau}{\sqrt{\chi^2 / (n_1 + n_2 - 2)}}$, where χ^2 is independent of τ and

has $df = (n_1 + n_2 - 2)$. As such, the quantity on the left hand side of the above equation is distributed as t with $df = (n_1 + n_2 - 2)$.

$$\text{A test for } H_0: \mu_1 - \mu_2 = \xi_0 \text{ is then given by the statistic } t = \frac{(\bar{x}_1 - \bar{x}_2) - \xi_0}{s' \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

with $df = (n_1 + n_2 - 2)$. This is called *Fisher's t*.

For acceptance or rejection of the hypothesis H_0 , one will have to compare the computed value of t' with the appropriate tabulated value, keeping in view the alternative hypothesis.

Following the usual procedure, it can be found that the confidence limits to

$(\mu_1 - \mu_2)$ are $(\bar{x}_1 - \bar{x}_2) - (t_{\alpha/2, n_1+n_2-2}) s' \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ and

$(\bar{x}_1 - \bar{x}_2) + (t_{\alpha/2, n_1+n_2-2}) s' \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$. Obviously, in both cases we are using s'^2 as

the estimate of the common variance σ^2 .

Next, consider the problem of testing a hypothesis regarding the ratio σ_1 / σ_2 or the problem of setting a confidence limit to the ratio. The difference between this problem and the corresponding problem mentioned in case II may be noted. Since μ_1 and μ_2 are unknown in the present case, they are replaced by their estimates $\bar{x}_1$ and $\bar{x}_2$, and we use the fact that

$$\frac{\sum_{j=1}^{n_1} (x_{1j} - \bar{x}_1)^2 / (n_1 - 1) \sigma_1^2}{\sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)^2 / (n_2 - 1) \sigma_2^2} = \frac{s_1'^2}{s_2'^2} \times \frac{\sigma_2^2}{\sigma_1^2} \text{ is an F with } (n_1 - 1, n_2 - 1) \text{ degrees of}$$

freedom. For testing $H_0: \sigma_1 / \sigma_2 = \xi_0$, we will use the F statistic, but now $F =$

$\frac{s_1'^2}{s_2'^2} \times \frac{1}{\xi_0^2}$ with $(n_1 - 1, n_2 - 1)$ degrees of freedom. The confidence limits for σ_1^2 / σ_2^2

now will be $\frac{1}{F_{\alpha/2; n_1-1, n_2-1}} \frac{s_1'^2}{s_2'^2}$ and $\frac{1}{F_{\alpha/2; n_2-1, n_1-1}} \frac{s_1'^2}{s_2'^2}$. The corresponding limits to

σ_1 / σ_2 will be the positive square roots of these quantities.

Check Your Progress 3

- 1) When is a significance test done using z? When is it done using t? What is different about the test for differences between proportions?

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- 2) The scores of random sample of 8 students on a physics test are given below. Test to see if the sample mean is significantly different from 65 at the .05 level

60, 62, 67, 69, 70, 72, 75, 80

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- 3) State the effect on the probability of a Type I and of a Type II error of:
- a) the difference between population means
 - b) the variance
 - c) the sample size
 - d) the significance level
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- 4) The following data are the life in hours of two batches of electric bulbs. Test whether there is a significant difference between the batches in respect of average length of life.
- Batch I: 1505, 1556, 1801, 1629, 1644, 1607, 1825, 1748.
- Batch II: 1799, 1618, 1604, 1655, 1708, 1675, 1728.
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31.7.3 Problems Relating to a Bivariate Normal Distribution

Suppose in a given population, the variables x and y are distributed in a bivariate normal form with means μ_x and μ_y , standard deviation σ_x and σ_y and correlation coefficient ρ . Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be random sample of size 'n' drawn from this distribution. We shall assume that all the parameters are unknown.

Test for correlation coefficient:

Here the sample correlation coefficient is
$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\left\{ \sum_i (x_i - \bar{x})^2 \right\}^{\frac{1}{2}} \left\{ \sum_i (y_i - \bar{y})^2 \right\}^{\frac{1}{2}}}$$

where $\bar{x}$ and $\bar{y}$ are the sample means. When $\rho = 0$, the sampling distribution of r assumes a simple form $f(r) = \frac{1}{B\left(\frac{1}{2}, \frac{n-2}{2}\right)} (1-r^2)^{(n-4)/2}$ and in that case

$r \sqrt{n-2} / \sqrt{1-r^2}$ is distributed as a t with $df = (n-2)$.

This fact provides us with a test for $H_0: \rho = 0$. As to the general hypothesis $H_0: \rho = \rho_0$, an exact test becomes difficult, because for $\rho \neq 0$ the sample correlation has a complicated sampling distribution. For moderately large n , there is an approximate test, which we have not discussed presently.

Problems regarding the difference between μ_x and μ_y :

Information regarding the difference between the means, μ_x and μ_y , may be of some importance when x and y are variables measured in the same units.

To begin with, we note that if we take a new variable, $z = x - y$, then this z , being a linear function of normal variables, is itself normally distributed with mean $\mu_z = \mu_x - \mu_y$ and variance $\sigma_z^2 = \sigma_x^2 + \sigma_y^2 - 2\rho\sigma_x\sigma_y$.

It will follow, from what we have said in the section for univariate normal distribution, that if we put $z_i = x_i - y_i$, $\bar{z} = \sum_i z_i / n$ and $s_z'^2 = \frac{1}{n-1} \sum_i (z_i - \bar{z})^2$, then $\sqrt{n}(\bar{z} - \mu_z) / s_z'$ will be distributed as a t with $df = (n - 1)$. This will provide us with a test for $H_0: \mu_x - \mu_y = \xi_0$, which is equivalent to $H_0: \mu_z = \xi_0$ and with confidence limits to the difference $\mu_z = \mu_x - \mu_y$. The statistic $\sqrt{n}(\bar{z} - \mu_z) / s_z'$ is often referred to as a *paired t*.

We may, instead, be interested in the ratio $\mu_x / \mu_y = \eta$ (say). In this case, we shall take $z = x - \eta y$, which again is normally distributed with mean $\mu_z = \mu_x - \eta\mu_y = 0$. Hence the statistic $t = \sqrt{n} \bar{z} / s_z'$ is distributed as at (i.e., *paired t*) with $df = (n - 1)$. This can be used for testing the hypothesis $H_0: \mu_x / \mu_y = \eta_0$ or for setting confidence limits for the ratio μ_x / μ_y .

Problems regarding the ratio σ_x / σ_y

When x and y are variables measured in identical units, one may also be interested in the ratio σ_x / σ_y . Let us denote the ratio by ξ .

If we consider the new variables $u = x + \xi y$ and $v = x - \xi y$,

then u and v are jointly normally distributed, like x and y , and

$$\text{cov}(u, v) = \sigma_x^2 - \xi^2 \sigma_y^2 = 0$$

Thus, u and v are uncorrelated normal variables.

In testing for the hypothesis $H_0: \sigma_x / \sigma_y = \xi_0$, we shall therefore take the new variables $u = x + \xi y$ and $v = x - \xi y$ and shall instead test for the equivalent hypothesis $H_0: \rho_{uv} = 0$. The test procedure for this will be given by the statistic $t = r_{uv} \sqrt{n-2} / \sqrt{1-r_{uv}^2}$ with $df = (n-2)$, r_{uv} being the sample correlation between u and v . To have confidence limits for ξ , we utilize the fact that, with $u = x + \xi y$ and

$$v = x - \xi y$$

$$P \left[\frac{|r_{uv}| \sqrt{n-2}}{\sqrt{1-r_{uv}^2}} \leq t_{\alpha/2, n-2} \right] = 1 - \alpha$$

$$\text{or, } P \left[\frac{r_{uv}^2 (n-2)}{1-r_{uv}^2} \leq t_{\alpha/2, n-2}^2 \right] = 1 - \alpha$$

By solving the equation $r_{uv}^2 (n-2) = t_{\alpha/2, n-2}^2 (1-r_{uv}^2)$ or, say, $\psi(\xi) = 0$ for the unknown ratio $\xi = \sigma_x / \sigma_y$, two roots will be obtained. In case, the roots, say ξ_1 and ξ_2 , are real ($\xi_1 < \xi_2$), this will be the required confidence limits for ξ with confidence coefficient $(1-\alpha)$.

Again, $\psi(\xi)$ may be either a convex or concave function. In the former case, we shall say $\xi_1 \leq \xi \leq \xi_2$, while in the latter we shall say $0 \leq \xi \leq \xi_1$ or $\xi_2 \leq \xi \leq \infty$.

But the roots may as well be imaginary, in which case we shall say that for the given sample $100(1-\alpha)\%$ confidence limits do not exist.

Check Your Progress 4

- 1) The correlation coefficient between the nasal length and stature for a group of 20 Indian adult males was found to be 0.203. Test whether there is any correlation between the characters in the population.

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- 2) a) What proportion of a normal distribution is within one standard deviation of the mean? b) What proportion is more than 1.8 standard deviations from the mean? c) What proportion is between 1 and 1.5 standard deviations above the mean?

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- 3) A test is normally distributed with a mean of 40 and a standard deviation of 7. a) What score would be needed to be in the 85th percentile? b) What score would be needed to be in the 22nd percentile?

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- 4) Assume a normal distribution with a mean of 90 and a standard deviation of 7. What limits would include the middle 65% of the cases?

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31.8 LET US SUM UP

In this unit, we have learnt how by using the test of significance, an estimator can be analysed and how sample observations can be tested for any statistical claim. The unit illustrates the way of testing various real-life problems. The basic concepts of hypothesis testing like null hypothesis, critical region, confidence limits, Type-I and Type-II errors are explained at length. Then we moved on to the understanding of the test procedure under normality assumption wherein we learnt the problems regarding the univariate normal distribution, comparison of two univariate normal distributions and problems relating to a bivariate normal distribution.

31.9 KEY WORDS

Alternative Hypothesis	: Any hypothesis, which is complementary to the null hypothesis, is called an alternative hypothesis, usually denoted by H_1 .
Confidence Interval and Confidence Limits	: If we choose once for all some small value of α (5% or 1%) i.e., level of significance and then determine two constants say, c_1 and c_2 such that $P [c_1 < \theta < c_2] = 1 - \alpha$, where θ is the unknown parameter then, the quantities c_1 and c_2 , so determined, are known as the 'confidence limits' and the interval $[c_1, c_2]$ within which the unknown value of the population parameter is expected to lie, is called the 'confidence interval' and $(1 - \alpha)$ is called the 'confidence coefficient'.
Critical Region	: A region (corresponding to a statistic t) in the sample space S , which amounts to rejection of H_0 , is termed as 'critical region'.
Level of Significance	: The probability that a random value of the statistic t belongs to the critical region is known as the 'level of significance'.
Null Hypothesis	: A definite hypothesis of no difference is called 'null hypothesis' and usually denoted by H_0 .
One -Tailed and Two - Tailed Tests	: A test of any statistical hypothesis where the alternative hypothesis is one-tailed (right-tailed or left-tailed) is called a 'one-tailed test'. For example, a test for testing the mean of a population $H_0: \mu = \mu_0$ against the alternative hypothesis: $H_1: \mu > \mu_0$ (right-tailed) or $H_1: \mu < \mu_0$ (left-tailed), is a 'one-tailed test'. Let us consider a test where the alternative

hypothesis is two-tailed like: $H_0: \mu = \mu_0$, against the alternative hypothesis $H_1: \mu \neq \mu_0$ ($\mu > \mu_0$ and $\mu < \mu_0$). This is known as a ‘two-tailed test’.

- Critical Values or Significant Values** : The value of the test statistic, which separates the critical (or rejection) region and the acceptance region is called the ‘critical value’ or ‘significant value’.
- Most Powerful Test** : The critical region W is the most powerful critical region of size α (and the corresponding test the most powerful test of level α) for testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ if $P(\mathbf{x} \in W | H_0) = \alpha$ and $P(\mathbf{x} \in W | H_1) \geq P(\mathbf{x} \in W_1 | H_1)$ for every other critical region W_1 satisfying the first condition.
- Parameter Space** : Let us consider a random variable x with p.d.f. $f(x, \theta)$. The p. d. f. of x can be written in the form $f(x, \theta)$, $\theta \in \Theta$. The set Θ , which is the set of all possible values of θ , is called the ‘parameter space’.
- Power of the Test** : The power of the test is defined as

$$\text{Power} = 1 - \text{Probability of Type II error}$$

$$= \text{Probability of rejecting } H_0 \text{ when } H_1 \text{ is true.}$$
- Type I and Type II Error** : Type I Error: Rejecting the null hypothesis H_0 when it is true.
 Type II Error: Accepting the null hypothesis H_0 when it is wrong, i.e., accept H_0 when H_1 is true
- Uniformly Most Powerful Test** : The critical region W is called the uniformly most powerful critical region of size α (and the corresponding test as uniformly most powerful test of level α) for testing $H_0: \theta = \theta_0$ against $H_1: \theta \neq \theta_0$. if $P(\mathbf{x} \in W | H_0) = \alpha$ and $P(\mathbf{x} \in W | H_1) \geq P(\mathbf{x} \in W_1 | H_1)$ for all $\theta \neq \theta_0$, whatever the other critical region W_1 satisfying the first condition may be.

31.10 ANSWER OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See Section 31.2
- 2) See Section 31.2
- 3) See Section 31.2